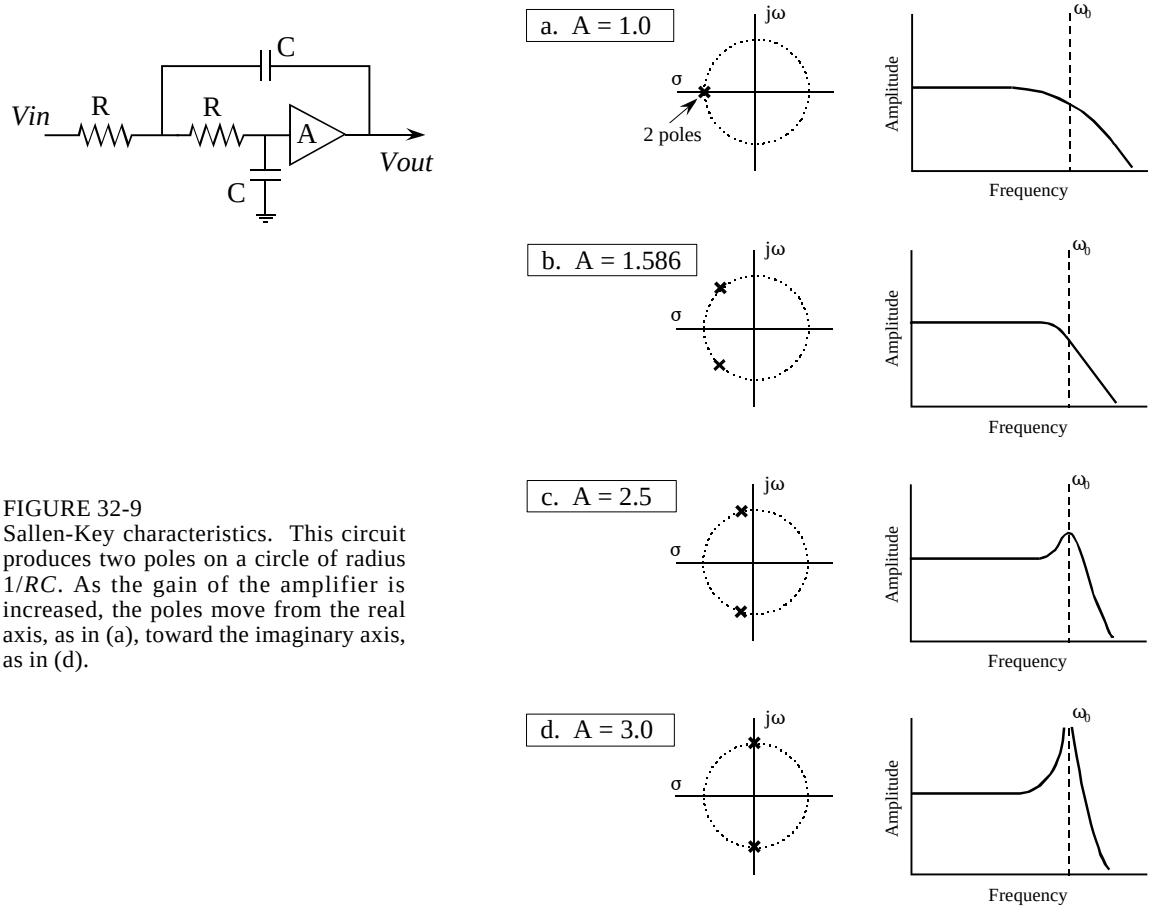

FIGURE 32-9
Sallen-Key characteristics. This circuit produces two poles on a circle of radius $1/RC$. As the gain of the amplifier is increased, the poles move from the real axis, as in (a), toward the imaginary axis, as in (d).

As the amplification is increased, the poles move along the circle, with a corresponding change in the frequency response. As shown in (b), an amplification of 1.586 places the poles at 45 degree angles, resulting in the frequency response having a sharper transition. Increasing the amplification further moves the poles even closer to the imaginary axis, resulting in the frequency response showing a peaked curve. This condition is illustrated in (c), where the amplification is set at 2.5. The amplitude of the peak continues to grow as the amplification is increased, until a gain of 3 is reached. As shown in (d), this is a special case that places the poles directly on the imaginary axis. The corresponding frequency response now has an infinity large value at the peak. In practical terms, this means the circuit has turned into an oscillator. Increasing the gain further pushes the poles deeper into the right half of the s-plane. As mentioned before, this correspond to the system being unstable (spontaneous oscillation).

Using the Sallen-Key circuit as a building block, a wide variety of filter types can be constructed. For example, a low-pass **Butterworth filter** is designed by placing a selected number of poles evenly around the left-half of the circle, as shown in Fig. 32-10. Each two poles in this configuration requires one